

SanctionX Project Report

AI-Powered Digital Loan Assistant

Comprehensive Technical & Market Analysis

1. Executive Summary

SanctionX is a state-of-the-art AI-powered digital loan officer designed to streamline the personal loan application process. By integrating advanced OCR, fraud detection, and automated credit assessment, it reduces the sanction time from days to minutes. The platform ensures RBI compliance while providing a premium, user-centric experience.

2. Technology Stack

Component	Technology	Description
Frontend	Next.js 16, React 19	Modern, high-performance UI framework
Backend	FastAPI (Python)	High-performance asynchronous API layer
Database	Supabase / PostgreSQL	Real-time data management with robust schema
Security	Firebase Auth / Supabase Auth	Secure user authentication and session management
OCR Engine	Tesseract / OpenCV	Automated extraction of data from documents
Logic/Agents	Custom Python Agents	Modular agents for Fraud, Eligibility, and KFS generation
Styling	Tailwind CSS 4	Custom-themed dark mode and premium aesthetics

3. Core Features

- **Smart Signup Wizard:** A 5-step multi-step form with real-time validation and auto-save.
- **AI Chat Assistant:** A conversational interface that guides users through the loan process.
- **Automated OCR:** Real-time extraction of income and PAN/Aadhaar details from uploaded documents.
- **Fraud Detection:** Integrated risk engine that checks for multiple PAN entries and income discrepancies.
- **Instant Sanction Letter:** Dynamic PDF generation of sanction letters and Key Fact Statements (KFS).
- **Admin Dashboard:** Comprehensive view for manual review and bulk application management.

4. Problem & Solution

The Problem:

Traditional loan processes are plagued by long wait times, excessive paperwork, and manual verification errors. Customers often face uncertainty during the approval process, leading to a poor user experience.

The SanctionX Solution:

SanctionX digitizes the entire journey. By using AI to verify identity and income instantly, it eliminates bottlenecks. The chat-based interface makes the process interactive and transparent, delivering a decision in under 5 minutes.

5. Market & Customer Dynamics

Target Audience:

Tech-savvy millennials, salaried professionals, and micro-entrepreneurs who require quick access to credit without the hassle of visiting a bank branch.

Market Opportunity:

The Digital Lending market in India is expected to reach \$350 Billion by 2026. SanctionX positions itself as a 'Digital Direct' platform, reducing customer acquisition costs for lenders by automating the top-of-the-funnel qualification.

6. Why This App is Needed

For Users: Speed, transparency, and a friction-less mobile-first experience. For Institutions: Lower operational costs, reduced fraud risk, and higher conversion rates through automated processing.